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# Comparison of Skyfield SGP4 and Basilisk Numerical Orbit Propagation for GNSS-R Formation-Flying Applications

*Student:*
Christoffer D. Hystad

*Supervisor:*
Jan Tommy Gravdahl

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# ABSTRACT

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# ABBREVIATIONS

CHAPTER

# ONE

# NOTES

Strongest narative for my thesis:

1. My specialization project showed that single-TLE SGP4 is not sufficient as ground truth, and that simplified drag limited the conclusions.
2. The Master's thesis improves upon the validation reference by using regularly update TLEs, making Skyfield-SGP4 a short-horizon, observation-updated reference
3. Basilisk is then validated as sufficiently accurate for long time-horizon mission simulation
4. The validated Basilisk propagator is used inside a realistic two-satellite GNSS-R formation simulator
5. Mission lifetime is estimated from fuel depletion + formation loss, with battery/power constraints shaping operations.

Scope: Focus on:

- Basilisk propagation validation
- Two-satellite deployment and formation acquisition
- Autonomous mode switching
- battery and fuel lifetime curves
- formation evolution
- mission lifetime estimate

Include, but keep on a high level:

- GNSS-R signal physics (just what is required to enable observation)
- Advanced controller derivations (both attitude and formation)

REMEMBER:

- Go over glosseries, and remove any that is only use once. Also make sure that 'gls' is used consistently for the same terms
- Use Inspera's 'plagiat' checker to verify that the theory sections ported from my specialization project report has been modified enough to be usable.
- In "Natural drift under Ideal two-body motion" I use the subscript "$L$" for the leader satellite, which has preveously been denoted using "$l$". I would just use the latter here too, but I cannot use "$f$" for the follower, since this denotes the RTN frame. Therefore: use $L$ for leader, and change the use throughout the text, INCLUDING the figures.
- In "Natural drift under Ideal two-body motion", the vector for orbital elements is denoted using $\boldsymbol{\alpha}$, which hasn't been declared previously. Add this in the "Classic orbtial elements" section.

CHAPTER

# TWO

# INTRODUCTION

## 2.1 Background & Motivation

Global Navigation Satellite Systems (GNSS) have become deeply embedded into our modern infrastructure with multiple constellations, such as GPS, GLONASS, Galileo and BeiDou, ensuring that four satellites are observable from the surface at any given time, thus providing worldwide coverage [**nasa_earthdata_gnss**]. The sheer number of operational satellites that broadcast GNSS signals makes it one of the most powerful signals of opportunity for remote sensing, enabling monitoring without the need for onboard transmitters.

At the same time, the space sector has recently undergone rapid changes. Launch costs have fallen significantly due to government outsourcing of launch providers to private companies, and the competitive environment it produced. This is clearly shown in figure **??**, where the per-kilogram launch cost has hit an all-time low with the deployment of SpaceX rockets *Falcon9* and *Falcon Heavy* [**jones2018launch_cost**]. As a result, it has become increasingly feasible for universities, research groups and smaller nations to deploy space-based systems dedicated to leveraging GNSS signals for remote sensing applications.

In parallel with this growth, there has also been an increase in the frequency of reported intentional GNSS interference to disrupt aviation operations. The interference can be divided into separate categories characterized by their difference in method and effect: *Jamming* refers to the relative high-power emission of Radio Frequency (RF) noise to block GNSS signals, while *spoofing* is the transmission of signals that resemble those from real satellites, misleading the GNSS receiver to think it has a different position [**ffi_factsheet_2024**]. The Norwegian Defense Research Establishment (FFI) states that: "In Norway's northernmost county of Finnmark, close to the Russian border, loss of GNSS signals due to jamming occurs nearly daily. [. . . ] The number of disruptions has increased significantly since Russia's invasion of Ukraine in 2022." [**ffi_jamming_2024**]. Data from *gpsjam.org* suggest that similar disruptions have been reported across Eastern

Europe, the Middle East, and the Arctic [**gpsjam_2025**]. These events highlight a broader geopolitical trend: GNSS interference is becoming more common. This places increased strategic value on systems that can detect and localize jamming and spoofing events.

GNSS Reflectometry (GNSS-R) offers a promising technique for addressing these challenges. According to Jin and Komjathy [**jin2010gnss-R**], Hall and Cordey [**hall_cordey_1988**] were the first to propose that GNSS L-band reflections could be exploited in a bistatic radar configuration to retrieve ocean surface properties, laying the foundation for modern GNSS-R techniques. Recent theory suggest that the same bistatic configuration can be used for surveillance of the GNSS spectrum, where multiple GNSS-R satellites in formation can detect and localize sources of jamming and spoofing. This emerging use-case is one of the central research goals of NTNU SmallSat Lab's GNSS-R project, which aim to develop methods for this process [**ntnu_smallsat_gnssr_2025**]. The project currently investigates this concept through its GNSS-R satellite mission study, exploring formations geometries, orbit configurations and system designs that would enable robust detection of GNSS interference sources and ships.

Accurate knowledge of spacecraft position and velocity is essential for the performance of any GNSS-R system. In such systems, regardless of application, the reflected GNSS signal is interpreted by comparing its time delay and Doppler shift relative to the direct signal. This process is highly sensitive to the specular reflection point's geolocation, which is given by the geometric relationship between the GNSS transmitter, the reflecting surface and the receiver. Even small errors in the satellite's propagated state shifts the estimated specular point by kilometers. Such errors can lead to misinterpretations of the delay-Doppler maps, for instance if the predicted specular point is over ocean when the true specular point occurs over land [**gleason2020cygnss**].

In real missions, the satellite orbit and relative geometry within satellite formations will naturally drift over time due to orbital perturbations. During mission design, simulation tools are used to predict the satellites motion. If the simulation does not accurately model the relevant perturbations, its predicted state evolution will diverge from physically realistic behavior, eventually yielding incorrect specular point locations and misleading conclusions about the performance of the GNSS-R system. Furthermore, inaccuracies in position and velocity directly translates to uncertainty in later analyses, like coverage analysis, or formation-keeping delta-V estimation. For these reasons, a high-fidelity propagator is necessary to ensure realistic and reliable evaluations of the GNSS-R system performance.

## 2.2 Problem Definition

A system simulator has already been developed to support the development of SmallSat Lab's GNSS-R project (cite???). The tool currently allows area coverage estimation and a detection probability analysis of a stationary maritime target, and is intended to be extended for ocean surveillance and GNSS Radio Frequency interference (RFI) detection in the future. As explored in the previous section, these analyses rely on accurately propagating both GNSS and GNSS-R satellites to minimize their uncertainties. Currently, the simulator uses the Simplified General Perturbations 4 (SGP4) propagator through the Skyfield Python library for this purpose.

The SGP4 propagator has long been the industry standard for orbit prediction, and has subsequently been the subject of thorough evaluation. Several studies document that the propagator can exhibit decreasing accuracy over longer time-horizons, particularly for formation-flight or precision geometry applications [**aida2013sgp4_accuracy**]. This raises concerns about whether the current SGP4-based implementation is sufficiently accurate for GNSS-R studies, which are highly sensitive to orbital state errors.

### 2.2.1 Research Questions

This motivates a systematic verification of the Skyfield-SGP4 implementation by comparing its long-term predictions against a propagator with higher modeling fidelity. Therefore, this specialization project will support the SmallSat Lab GNSS-R mission by developing such a propagator within the Basilisk simulation framework, which natively supports detailed perturbation models and allows for controlled comparisons with Skyfield-SGP4. The study uses this capability to evaluate the differences between the propagators and evaluate their implications for GNSS-R mission analysis. As a result, the project is guided by three research questions:

- a

Answering these questions will support the project's goal of guaranteeing the reliability and accuracy of orbital propagation tools used for GNSS-R small-satellite formation studies in future mission design.

## 2.3 Overall Method

## 2.4 Scope & Limitations

This specialization project focuses on evaluating the long-term orbital propagation accuracy of two distinct simulation frameworks, Skyfield-SGP4 and a high-fidelity Basilisk propagator, in the context of GNSS-R formation-flying missions. To ensure a controlled and way defined comparison, the project's scope has been limited to a set of simulation configurations and perturbation models.

### 2.4.1 Simulation Configuration Scope

The analysis covers a relatively long time-horizon, ranging from several days up to one week, in order to adequately capture the frameworks' divergence over a mission-relevant time span. The simulated formation consists of two satellites flying in Low Earth Orbit (LEO) in a leader-follower configuration. Both are initialized on the same orbital plane with nominally constant along-track separation and with the same initial conditions to guarantee same-case comparisons.

### 2.4.2 Limitations

Several aspects of GNSS-R mission analysis lie outside the scope of this specialization project. First, no attitude dynamics are modeled. The satellites are assumed to maintain a fixed nadir-oriented attitude or otherwise idealized pointing, which removes attitude-orbit coupling and optical or RF pointing constraints from consideration.

Second, the project includes no GNSS-R signal modeling or bistatic radar simulation. The study evaluates orbital geometry only; signal processing elements such as delay-Doppler map generation, specular-point reflectivity, link budget modeling, or interference detection algorithms fall beyond the project's objectives.

Third, no active formation-keeping control is implemented. The satellites follow purely ballistic trajectories under the applied perturbations. As such, the delta-V analysis is limited to estimating how often and how strongly formation drift would need to be corrected, rather than simulating actual guidance, navigation, and control (GNC) actions.

Finally, conclusions drawn from this study are limited to the tested scenarios. The orbital regimes, perturbation models, and formation geometry considered here may not generalize to all LEO missions, nor to GNSS-R concepts employing different formation layouts, attitude profiles, or operational constraints.

These limitations naturally define the boundary between the present specialization project and a potential master's thesis. While this project focuses on verifying orbital propagation fidelity, a master's thesis could extend the analysis by incorporating full attitude modeling, GNSS-R signal simulation, operational formation-keeping strategies, and end-to-end detection performance. Such expansions would build upon the validated propagation models developed here, enabling a more comprehensive assessment of GNSS-R formation-flying missions.

## 2.5 Report Structure

Following is a brief overview of chapters in this report, and what each of them cover.

CHAPTER

# THREE

# LITERATURE REVIEW

The literature review conducted for this paper can be divided into three themes:

- GNSS-R Small-Satellite Missions
- Formation Flying and Formation Maintenance for Small Satellites
- Integrated Satellite Mission Simulation and Lifetime Analysis

## 3.1 GNSS-R Small-Satellite Missions

- Scientific evolution:
    1. GNSS as a signal of opportunity
    2. GNSS-R feasibility for different remote sensing applications. Bistatic radar principle
    3. Early demonstrations of GNSS-R remote sensing on aircraft mission
    4. Spacecraft demonstrations
    5. Validation of technology for different applications
    6. Expansion of GNSS-R Mission applications (include jamming/spoofing detection)
- Mapping useful GNSS-R missions after validation of technology at spacecraft altitudes
- Feasibility of cooperative swarms and their respective formation types w. benefits

Reflected GNSS signals were first discussed as a

## 3.2 Formation Flying and Formation Maintenance for Small Satellites

## 3.3 Integrated Satellite Mission Simulation and Lifetime Analysis

## 3.4 Research gaps and thesis contributions

CHAPTER

# FOUR

# BACKGROUND THEORY

## 4.1 Reference Frames

A reference frame $\mathcal{F}^{rf} : \{\mathcal{O}_{rf};\ \hat{\mathbf{x}}_{rf}, \hat{\mathbf{y}}_{rf}, \hat{\mathbf{z}}_{rf}\}$ is uniquely defined by three dextral orthogonal unit vectors $\{\hat{\mathbf{x}}_{rf}, \hat{\mathbf{y}}_{rf}, \hat{\mathbf{z}}_{rf}\}$ and the origin placement $\mathcal{O}_{rf}$ [1]. This section will present the definitions for the most important reference frames used in the thesis. The reader should note that the individual reference frame subscripts are assigned to be consistent with the conventional Basilisk notation.

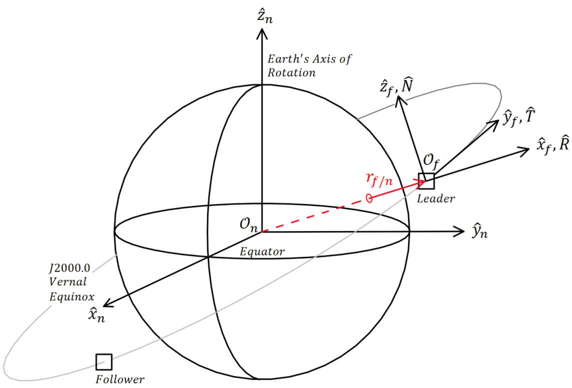


**Figure 4.1.1:** Illustration of the ECI ($\mathcal{F}^n$)and RTN ($\mathcal{F}^f$) reference frame definitions. The basis unit vectors are displayed with different lengths for illustrative purposes. Adopted from [2, 3] and modified from [4]

.

### 4.1.1 J2000 Earth-Centered Inertial (ECI) Frame

The ECI reference frame, $\mathcal{F}^n : \{\mathcal{O}_n;\ \hat{\mathbf{x}}_n, \hat{\mathbf{y}}_n, \hat{\mathbf{z}}_n\}$, is fixed relative to the distant stars, and is therefore commonly approximated as an inertial frame in which Newtonian mechanics are valid [5]. The frame is defined with its origin in the Earth's Center of Mass (CM), with $\hat{\mathbf{z}}_n$ pointing along its rotational axis towards the geographical North Pole, $\hat{\mathbf{x}}_n$ pointing towards the vernal equinox through the equator, and $\hat{\mathbf{y}}_n$ defined according to the right-hand rule. Multiple variants of the ECI definition exists, primarily varying in the epoch used to define the Earth's equatorial plane and equinox direction. In this thesis, the J2000 ECI frame is adopted. It is defined using the J2000.0 reference epoch, equivalent to 1st January 2000, at 12:00 Terrestrial Time (TT) [6]. The frame definition is illustrated in Figure 4.1.1.

### 4.1.2 Earth-Centered Earth-Fixed (ECEF) Frame

The Earth-Centered Earth-Fixed (ECEF) reference frame $\mathcal{F}^e : \{\mathcal{O}_e;\ \hat{\mathbf{x}}_e, \hat{\mathbf{y}}_e, \hat{\mathbf{z}}_e\}$ is fixed relative to the Earth's surface, with the frame origin positioned in its CM. This thesis adopts the WGS84 ECEF formulation as presented by Stryjewski in [7], where the $\hat{\mathbf{x}}_e$ axis points through the Prime Meridian in the equatorial plane, $\hat{\mathbf{z}}_e = \hat{\mathbf{z}}_n$ points along the Earth's rotational axis, and $\hat{\mathbf{y}}_e$ is defined according to the right-hand rule. Because of the Earth's angular velocity, the transformation $\mathbf{R}_n^e$ between ECI to ECEF can be expressed using a time-dependent rotation angle $\theta(t)$ about the shared $\hat{\mathbf{z}}_e$ axis [4]

$$\mathbf{R}_n^e = \begin{bmatrix} \cos\theta(t) & -\sin\theta(t) & 0 \\ \sin\theta(t) & \cos\theta(t) & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{R}_e^n = (\mathbf{R}_i^e)^T \tag{4.1}$$

### 4.1.3 Radial-Transverse-Normal (RTN) Frame

The Radial-Transverse-Normal (RTN) reference frame, $\mathcal{F}^f : \left\{\mathcal{O}_f;\ \hat{\mathbf{x}}_f, \hat{\mathbf{y}}_f, \hat{\mathbf{z}}_f\right\}$, is a local orbital frame that is useful for expressing relative motion between spaceborne objects. Vallado presents the frame as the *Satellite coordinate system, RSW* in [8], but the *RTN* name is chosen for this thesis to better convey each component's physical significance. For use in formation flight, the coordinate frame origin is assigned to the leader satellite, $l$'s CM. The dextral orthogonal unit vectors are defined with the $\hat{\mathbf{x}}_f$ axis pointing in the radial direction along $\mathbf{r}_{l/n}^n$, the $\hat{\mathbf{z}}_f$ axis points normal to the orbital plane, and $\hat{\mathbf{y}}_f$ fulfills the right-hand rule. Note that the latter will always point in direction of positive velocity, but will only be perpendicular to the velocity for circular orbits [8, 9]. Using this definition, the frame can be expressed mathematically as:

$$\hat{\mathbf{R}}^n = \hat{\mathbf{x}}_f^n = \frac{\mathbf{r}_{l/n}^n}{||\mathbf{r}_{l/n}^n||}, \qquad \hat{\mathbf{N}}^n = \hat{\mathbf{z}}_f^n = \frac{\mathbf{r}_{l/n}^n \times \mathbf{v}_{l/n}^n}{||\mathbf{r}_{l/n}^n \times \mathbf{v}_{l/n}^n||}, \qquad \hat{\mathbf{T}}^n = \hat{\mathbf{y}}_f^n = \hat{\mathbf{z}}_f^n \times \hat{\mathbf{x}}_f^n \tag{4.2}$$

Here, $\mathbf{r}_{l/n}^n$ and $\mathbf{v}_{l/n}^n$ are used to denote the Earth-relative leader position and velocity expressed in ECI, while $\hat{\mathbf{y}}_f$ and $\hat{\mathbf{z}}_f$ correspond to the along- and cross-track directions, respectively. An illustration of the leader satellite's position relative to $\mathcal{O}_n$ and its affixed RTN frame is shown in Figure 4.1.1.

### 4.1.4 Body-Fixed Frame

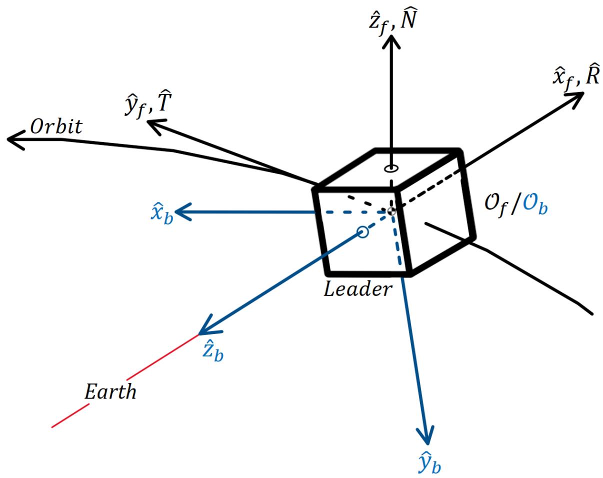


**Figure 4.1.2:** Illustration of the body-fixed ($\mathcal{F}^b$)and RTN ($\mathcal{F}^f$) reference frame definitions for a 1U CubeSat. Adopted from [10].

The body-fixed reference frame, $\mathcal{F}^b : \{\mathcal{O}_b;\ \hat{\mathbf{x}}_b, \hat{\mathbf{y}}_b, \hat{\mathbf{z}}_b\}$, is a moving reference frame that is rigidly attached to the spacecraft body. Because of this property, the body-fixed frame is used for expressing satellite attitude relative to an inertial frame, actuator torques and sensor measurements, since these quantities are naturally expressed relative to the body itself. The orientation of its axes are typically chosen to best represent the spacecraft geometry, and to coincide with the principal axes of inertia [11]. Consequentially, CubeSat body axes are commonly selected to be perpendicular to its side panels. The frame origin is commonly placed at the CM in order to simplify the rotational equations of motion, and to reduce translational-rotational coupling [12]. An illustration of the body frame definition for a 1U CubeSat is shown in Figure 4.1.2.

## 4.2 Modified Rodrigues parameters

Modified Rodrigues Parameters (MRP)s is a 3-element attitude parameterization commonly used for spacecraft attitude control. The representation can be derived from the Euler-Rodrigues symmetric parameters, better known as the unit quaternion:

$$\mathbf{q} = \begin{bmatrix} \boldsymbol{\epsilon} \\ \eta \end{bmatrix} = \begin{bmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 & \eta \end{bmatrix}^T, \qquad \mathbf{q}^T\mathbf{q} = |\boldsymbol{\epsilon}|^2 + \eta^2 = 1 \tag{4.3}$$

where $\eta$ is the scalar component and $\boldsymbol{\epsilon}$ is the imaginary vector. Using this definition, Shuster writes in [13] that the MRPs are obtained through a stereographic projection of the unit quaternion onto $\mathbb{R}^3$

$$\mathbf{p} = \frac{\boldsymbol{\epsilon}}{1+\eta}, \qquad \mathbf{m} = \frac{\boldsymbol{\epsilon}}{1-\eta} \tag{4.4}$$

Here, $\mathbf{p}$ and $\mathbf{m}$ are the positive and negative forms, differing only in the singularity placement. This thesis utilizes the positive form, and will hereby be referred to as $\boldsymbol{\sigma}$. Using the axis-angle representation of a rotation, where $\hat{\mathbf{n}}$ represents the unit rotation axis and $\theta$ is the angle of rotation about the axis, the positive-form MRPs can be expressed as:

$$\mathbf{q} = \begin{bmatrix} \hat{\mathbf{n}} \sin(\theta/2) \\ \cos(\theta/2) \end{bmatrix} \quad \Rightarrow \quad \boldsymbol{\sigma}(\theta, \hat{\mathbf{n}}) = \hat{\mathbf{n}} \tan\left(\frac{\theta}{4}\right), \tag{4.5}$$

From this formulation, it follows that the MRPs become singular as $\theta \to 2\pi$, since

$$\tan\left(\frac{\theta}{4}\right) \to \infty \quad \text{for} \quad \theta \to 2\pi \tag{4.6}$$

To avoid numerical issues near the singularity, an equivalent MRP shadow set, $\boldsymbol{\sigma}^s$, may be used [14]. The shadow set transformation is defined as:

$$\boldsymbol{\sigma}^s = -\frac{\boldsymbol{\sigma}}{\boldsymbol{\sigma}^T \boldsymbol{\sigma}} \tag{4.7}$$

Switching to the shadow set when $||\boldsymbol{\sigma}|| > 1$ effectively bounds the MRP set by the unit sphere, and prevents the parameterization from approaching the singularity numerically [15, 14, 13].

## 4.3 Classic Orbital elements

The classic orbital elements, also known as Keplerian elements, are a set of six parameters used to uniquely define the shape, size and orientation of an orbital plane, as well as the position of an orbiting body [16, 8]. In astrodynamics, these parameters provide a compact and intuitive representation of orbital motion, commonly used for orbital analyses and spacecraft guidance. The classic orbital elements used in this thesis are:

$$\left.\begin{array}{ll} a: & \text{semimajor axis} \\ e: & \text{eccentricity} \end{array}\right\} \quad \text{orbital plane size and shape}$$

$$\left.\begin{array}{ll} i: & \text{inclination} \\ \Omega: & \text{right ascension of the ascending node (RAAN)} \\ \omega: & \text{argument of perigee} \end{array}\right\} \quad \text{orbital plane orientation}$$

$$\left.\begin{array}{ll} M: & \text{mean anomaly} \end{array}\right\} \quad \text{orbital body position}$$

It is important to note that the semimajor axis, and mean anomaly are in some orbital element configurations replaced with the angular momentum $h$, and the true anomaly $\theta$, respectively. In any case, these six parameters define a Keplerian orbit, and they can be grouped according to the orbital property they describe: The semimajor axis and eccentricity define the shape and size of the orbital plane, while the inclination, Right Ascension of the Ascending Node (RAAN) and argument of perigee describe the plane's orientation. Finally, the mean anomaly is used to describe the spaceborne object's position along the orbital trajectory. An illustration of the classic orbital elements is shown in Figure 4.3.1.

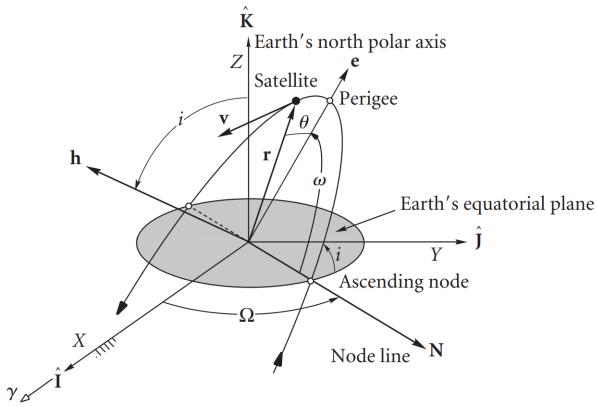


**Figure 4.3.1:** Orbital elements relative to the ECI frame. Courtesy of [16]

.

The semimajor axis is defined as half of the longest ellipse diameter, and determines the overall size of the orbit. For elliptical orbits, this parameter is directly coupled with the orbital period through Kepler's third law. The eccentricity is a measure of how much the orbital trajectory deviates from a circle. It takes values in the span between zero and one, where an eccentricity of zero corresponds to a perfectly circular orbit, and an increasing eccentricity correspond to an elliptical orbit of increasing elongation.

The inclination is the intersection angle between the orbital plane and the reference plane, commonly the equatorial plane spanned by the $\hat{x}_n$ and $\hat{y}_n$ axes in ECI. This angle determines the tilt of the orbital plane, while its orientation about the $\hat{z}_n$ axis is determined by the RAAN parameter. For nonzero inclinations, the orbital trajectory intersects the reference plane at two points, the ascending and descending nodes. The ascending node is where the object travels from the Southern to the Northern Hemisphere, and RAAN is defined as the angle between this node and the $\hat{x}_n$ axis in the equatorial plane.

The argument of perigee specifies the orientation of the ellipse within the orbital plane. It is defined as the angle from the ascending node to the perigee along the orbital trajectory, where perigee is a term specific to Earth satellites, describing the point along the trajectory closest to Earth. Finally, the mean anomaly defines the orbiting body's position along the orbit as a function of time. It is defined as the elapsed fraction of the orbital period since the orbiting body passed the perigee [16, 17].

## 4.4 Orbital Perturbations

D.A. Vallado defines orbital perturbations as deviations from the ideal Keplerian two-body motion caused by additional gravitational or non-gravitational accelerations acting on a spacecraft [8]. For an Earth-orbiting satellite, the Earth's oblateness, atmospheric drag, Solar Radiation Pressure (SRP) and third-body attraction from the Sun and Moon all cause its trajectory to deviate from the ideal point-mass solution. The following subsections will therefore address each major perturbation effect in approximate decreasing order of significance for LEO satellites in accordance with Montenbruck & Gill in [18].

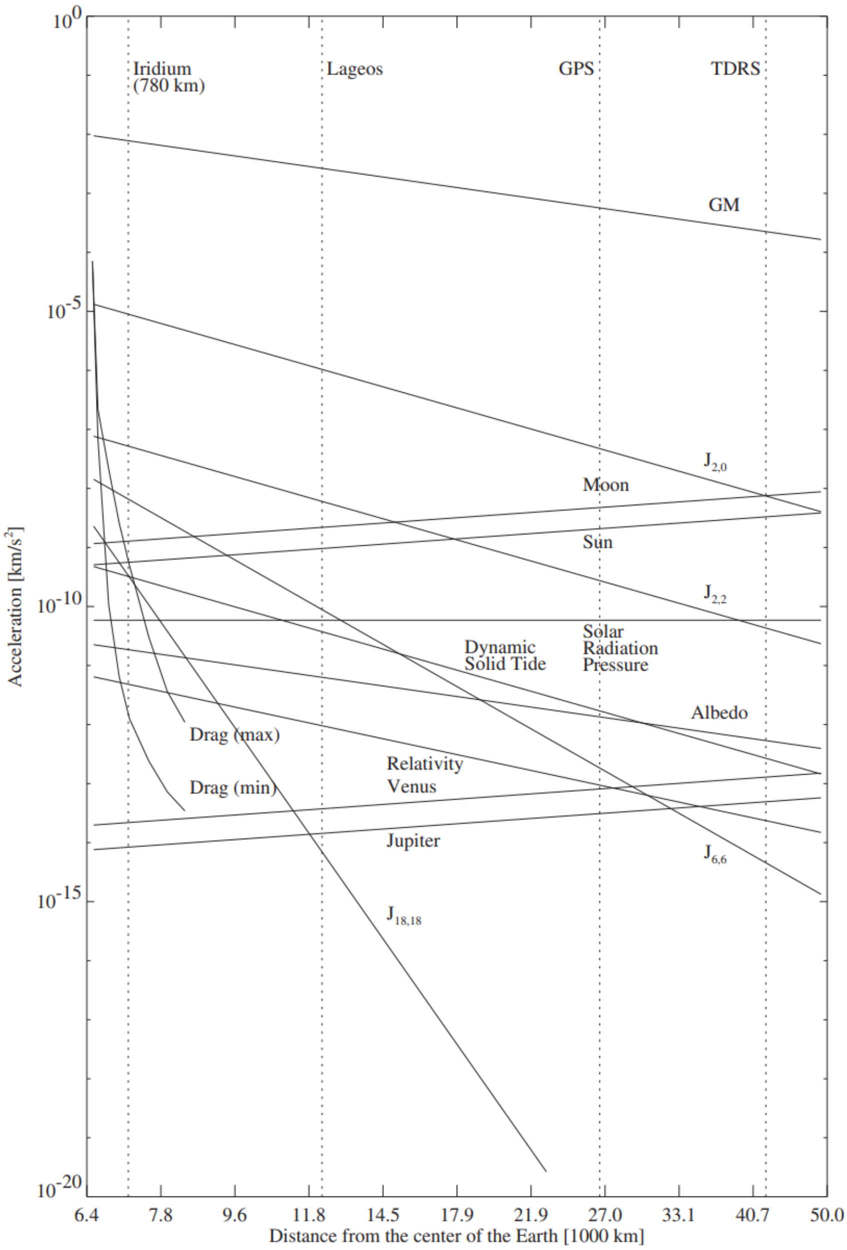


**Figure 4.4.1:** Acceleration magnitudes caused by various orbital perturbations. Courtesy of [18]

.

### 4.4.1 Earth's Oblateness

Keplerian motion assumes that the Earth can be represented as a perfectly spherical body with all its mass concentrated in the earth-centered inertial frame origin. Under this assumption, the gravitational force acting on a spacecraft, $s$, can be modeled using Newton's law of gravity

$$\mathbf{a}_{grav} = \left(\ddot{\mathbf{r}}_{s/n}\right)_{grav} = -\frac{\mu}{||\mathbf{r}_{s/n}||_2^3}\mathbf{r}_{s/n} \iff \left(\ddot{\mathbf{r}}_{s/n}\right)_{grav} = \nabla U, \quad \text{where } U = \mu\frac{1}{||\mathbf{r}_{s/n}||_2} \tag{4.8}$$

Using the notation from [4], $\mu = GM$ is the Earth's gravitational constant, and $\nabla U$ is the gradient of the gravity potential. In accordance with the spherical assumption, the gravity potential is only dependent on the satellite's geocentric distance, implying uniform density. In actuality, the Earth's mass is unhomogeneously distributed through its volume, and centrifugal forces causes the planet to bulge outwards around the equator due to its angular velocity. These irregularities produce perturbing accelerations that gradually alter the orbital trajectories over time.

Montenbruck & Gill explain that these deviations from spherical symmetry can be modeled by expanding the Earth's gravity potential into a spherical harmonic series representation [18]. The resulting series expresses the geopotential as a weighted sum of spherical harmonic terms, describing gravitational irregularities caused by the Earth's unhomogeneous mass distribution. Here, the spacecraft position is represented in ECEF spherical coordinates consisting of the geocentric distance $r = ||\mathbf{r}^e_{s/n}||_2$, the geocentric latitude $\phi^e$, and the longitude $\lambda^e$. Using these coordinates together with the associated Legendre polynomials, the geopotential may be written in the general spherical harmonic form as

$$U(r, \phi^e, \lambda^e) = \frac{\mu}{R}\sum_{n=0}^{\infty}\sum_{m=0}^{n}\left(C_{nm}V_{nm} + S_{nm}W_{nm}\right) \tag{4.9}$$

where,

$$V_{nm} = \left(\frac{R}{r}\right)^{n+1} P_{nm}(\sin\phi^e)\cos(m\lambda^e), \qquad W_{nm} = \left(\frac{R}{r}\right)^{n+1} P_{nm}(\sin\phi^e)\sin(m\lambda^e). \tag{4.10}$$

Here, $R$ denotes the Earth's radius, $P_{nm}$ are the associated Legendre polynomials, while $C_{nm}$ and $S_{nm}$ are the normalized geopotential coefficients that encode the Earth's non-uniform mass distribution. The $_{nm}$ subscript denotes the degree $n$ and order $m$, where increasing value correspond to decreasingly significant perturbation terms. The harmonic degree relates to longitudinal mass variation, while the order determines the longitudinal dependency on the harmonic term.

The reader should note that a common simplification is to assume order $m = 0$ due to comparatively small variations in the Earth's gravity field along the longitude. This yields the zonal harmonic expression, where the $S_{n0}$ terms vanish, and the geopotential coefficients are commonly expressed as $J_n = -C_{n0}$. This gives the following expression derived from [18, 8]:

$$U(r, \phi^e) = \frac{\mu}{r}\left[1 - \sum_{n=2}^{4} J_n\left(\frac{R}{r}\right)^n P_n(\sin\phi^e)\right] \tag{4.11}$$

The resulting zonal model captures the dominant long-term perturbation effects while significantly reducing the complexity of the gravity model. Amongst the zonal coefficients, the second degree term $J_2$ dominates the non-spherical effects, as is clearly evident by the substantially larger acceleration magnitude of $J_{2,0}$ in Figure 4.4.1 compared to the other terms. The Figure also illustrates that the higher degree and order terms rapidly decrease in magnitude. However, to include the dominant lateral and longitudinal non-spherical effect, this thesis employs the general spherical harmonic formulation in Equation 4.9, with truncated degree and order.

### 4.4.2 Atmospheric Drag

Orbital atmospheric drag is caused by a momentum exchange between the orbiting body's surface and colliding atmospheric molecules along its trajectory. According to Montenbruck & Gill, this effect is the dominant non-gravitational perturbation acting on a spacecraft in LEO, as is also illustrated in Figure 4.4.1. Since the drag force acts opposite to the spacecraft's velocity, it continuously removes orbital energy, and causes a gradual reduction in orbital altitude and velocity over time [8, 18, 4].

For spacecraft operating in LEO, the interaction between the spacecraft surface and surrounding atmosphere is commonly modeled using the free-molecular flow assumption. In this regime, the atmospheric density is sufficiently low such that the inter-molecular collisions are negligible compared to the molecule-surface interaction. The drag acceleration experienced by a spacecraft $s$ can then be expressed using the classical aerodynamic formulation, as presented in [18]

$$\mathbf{a}_{\mathrm{D}}^{n} = \left(\ddot{\mathbf{r}}_{s/n}^{n}\right)_{D} = -\tfrac{1}{2} C_D \, \frac{A_D}{m_s} \, \rho \, v_{\mathrm{rel}}^2 \, \hat{\mathbf{v}}_{rel}^{n} \tag{4.12}$$

where $C_D$ is the dimensionless drag coefficient, $A_D$ is the effective cross-sectional area normal to the incoming flow, $m_s$ is the spacecraft mass and $\rho$ is the local atmospheric density. $v_{\mathrm{rel}}$ denotes the relative speed, while $\hat{\mathbf{v}}_{rel}^{n}$ is the unit vector in the direction of relative velocity. It is important to note that the relative velocity differs from the inertial spacecraft velocity because it represents its velocity relative to the local atmosphere [4]. Following Montenbruck & Gill in [18], the relative velocity can therefore be approximated by assuming the atmosphere co-rotates with the Earth

$$\mathbf{v}_{rel}^{n} = \mathbf{v}_{s/n}^{n} - \boldsymbol{\omega}_{e/n}^{n} \times \mathbf{r}_{s/n}^{n} \tag{4.13}$$

where $\{\mathbf{r}_{s/n}^{n}, \mathbf{v}_{s/n}^{n}\}$ are the spacecraft position and velocity expressed in the ECI frame, and $\boldsymbol{\omega}_{e/n}^{n}$ is the Earth's angular velocity vector. Amongst the parameters in the classic drag formulation, the atmospheric density $\rho$ is generally the largest source of modeling uncertainty. In this thesis, two density models will be evaluated: the NRLMSISE-00 model and the exponential density model. The exponential density model is a heavily simplified alternative, and assumes exponentially decreasing density with increasing altitude.

$$\rho(h) = \rho_0 \exp\left(-\frac{h - h_0}{H_0}\right), \qquad H_0 = \frac{\mathcal{R}T}{\mu_m g} \tag{4.14}$$

Here, $\rho_0$ is the reference density at some geocentric reference height $h_0$, and $h$ is the spacecraft altitude above the reference height. Finally, $H_0$ is the density scale height, whose equation in 4.14 can be derived from the hydrostatic equation and gas law from thermodynamics. This model captures the approximate atmospheric decay with altitude, but fails to account for major spatial and temporal effects, such as solar activity, geomagnetic disturbances, longitudinal/lateral variations and long-term solar cycles [18].

The NRLMSISE-00 empirical model accounts for these spatial and temporal atmospheric variations, and has been developed using several decades of atmospheric observations and satellite measurements. In addition to the density, the model also estimates other atmospheric parameters, such as temperature and individual molecule densities. Because of the spatial and temporal dependency of these parameters, the empirical model requires additional inputs describing both the spacecraft environment and current atmosphere conditions. These include the spacecraft altitude, like the exponential density model, but also the geographic location, current time and different time-averaged measurements of solar activity represented through the $F_{10.7}$ solar radio flux, and the geomagnetic index $A_p$ [19]. Together, these parameters allow the model to account for short- and long-term variations in the atmospheric density.

### 4.4.3 Solar radiation pressure

Like atmospheric drag, SRP is a non-gravitational perturbation caused by momentum transfer between the spacecraft and incident particles. However, in the case of SRP, this transfer originates from photons rather than residual atmospheric molecules. When photons are absorbed or reflected by the spacecraft surface, they transfer their momentum to the spacecraft, generating a small force in the approximate direction away from the Sun. Because the force acts continuously over the illuminated surface area, it is referred to as a radiation pressure [18, 4].

As discussed by Montenbruck & Gill in [18], the radiation pressure does not vary with the geocentric altitude directly, but rather the distance from the Sun according to the inverse-square law. Because the variation in Sun-spacecraft distance is negligible compared to the Earth orbital altitude, the resulting SRP acceleration experienced by an Earth-orbiting spacecraft is approximately constant, as illustrated in Figure 4.4.1. Although the SRP acceleration remains approximately constant with altitude, its relative significance increases at higher altitudes where atmospheric drag becomes progressively weaker, especially for spacecraft with large area-to-mass ratios.

For a flat spacecraft surface, the resulting SRP acceleration depends on the relative geometry between the solar radiation direction $\mathbf{r}^n_{Sun/s}$, and the spacecraft surface normal $\hat{\mathbf{n}}$. Using the formulation presented by Montenbruck & Gill, the acceleration in the ECI frame can be written as

$$\mathbf{a}^n_R = \left(\ddot{\mathbf{r}}^n_{s/n}\right)_R = -P_{Sun}\frac{\mathrm{AU}^2}{||\mathbf{r}^n_{Sun/s}||_2^2}\frac{A_R}{m_s}\cos(\theta)\left[(1-\varepsilon)\hat{\mathbf{r}}^n_{Sun/s} + 2\varepsilon\cos(\theta)\hat{\mathbf{n}}_s\right] \tag{4.15}$$

where $P_{Sun}$ denotes the solar radiation pressure at one astronomical unit ($AU$), the parameter $A_R/m_s$ represents the spacecraft sun-normal cross-sectional area-to-mass ratio, and $\theta$ is the angle between the incident radiation direction and $\hat{\mathbf{n}}$. Finally, $\varepsilon$ denotes to the fraction of reflected incoming radiation, where $\varepsilon = 0$ and $\varepsilon = 1$ correspond to perfect absorption and reflection, respectively.

For many orbit propagation applications, Montenbruck & Gill further argue that the simplified assumption $\theta = 0$ provides sufficient accuracy[18]. Under this assumption, the SRP model reduces to the commonly used cannonball model

$$\mathbf{a}_R^n = \left(\ddot{\mathbf{r}}_{s/n}^n\right)_R = -P_{Sun} C_R \frac{A_R}{m_s} \frac{\mathbf{r}_{Sun/s}^n}{||\mathbf{r}_{Sun/s}^n||_2^3} \mathrm{AU}^2 \tag{4.16}$$

where $C_R$ is the radiation pressure coefficient. This approximation models the spacecraft as an idealized spherical body with uniform reflective properties, thus simplifying the SRP model while keeping the dominant perturbation behaviour [8, 4].

### 4.4.4 Third-Body Perturbations

In addition to the Earth's gravitational field, Earth-orbiting spacecraft are subjected to gravitational attraction from other massive bodies within the solar system. For satellites orbiting around the Earth, the Sun and Moon are the dominant sources of third-body perturbation. Although these bodies are located far from the spacecraft compared to the Earth, their gravitational influence still produce measurable orbital perturbations, particularly for satellites operating in Medium Earth Orbit (MEO), Geostationary Earth Orbit (GEO), or highly elliptical trajectories, as presented by Figure 4.4.1 [18, 4].

Third-body perturbations introduce additional gravitational accelerations that continuously vary with the relative geometry between the spacecraft, Earth and the perturbing body. Although the perturbing accelerations from the Sun and Moon are much smaller than the Earth's central gravitational attraction, Montenbruck & Gill note that the influence become increasingly significant with orbital altitude, becoming comparable to geopotential perturbations of low degree [18]. Because the perturbing acceleration changes direction throughout the orbit, the resulting effect manifests as gradual deviations from ideal Keplerian motion, and contributes both periodic and long-term variations in the spacecraft trajectory, which is consistent with the general three-body dynamics described by Curtis in [16].

Following the formulation presented by Montenbruck & Gill, the perturbing acceleration from a third massive body $B$ is obtained by subtracting the gravitational acceleration excerted by the perturbing body on the Earth from the gravitational acceleration acting on the spacecraft. Using the same notation as in [4], the perturbing acceleration expressed in the ECI frame becomes

$$\mathbf{a}_B^n = \left(\ddot{\mathbf{r}}_{s/n}^n\right)_B = \mu_B \left( \frac{\mathbf{r}_{B/n}^n - \mathbf{r}_{s/n}^n}{||\mathbf{r}_{B/n}^n - \mathbf{r}_{s/n}^n||_2^3} - \frac{\mathbf{r}_{B/n}^n}{||\mathbf{r}_{B/n}^n||_2^3} \right) = \mu_B \left( \frac{\mathbf{r}_{B/s}^n}{||\mathbf{r}_{B/s}^n||_2^3} - \frac{\mathbf{r}_{B/n}^n}{||\mathbf{r}_{B/n}^n||_2^3} \right) \tag{4.17}$$

When multiple perturbing bodies are considered simultaneously, the total third-body acceleration is obtained by summing the individual contributions from each body. Using the major perturbing bodies for a LEO satellite, this becomes

$$\mathbf{a}_{B,\mathrm{tot}}^n = \mathbf{a}_{\mathrm{Sun}}^n + \mathbf{a}_{\mathrm{Moon}}^n \tag{4.18}$$

## 4.5 Differential Perturbations in Formation-Flying

Alfriend et al. refers to the generally agreed upon definition of spacecraft formation flying in [20] as: "The tracking or maintenance of a desired relative separation, orientation or position between or among spacecraft". It then follows that differential perturbations describe the relative effect of environmental accelerations acting on two satellites in formation, that cause the formation to naturally drift away from its desired geometry over time. The difference in experienced accelerations by the two spacecraft in formation can arise because the spacecraft occupy slightly different obits, positions, velocities and attitudes, or because they have different physical attributes. In formation-flying, even small prolonged differences in experienced acceleration can accumulate over many orbital periods to gradually change the radial, along- or cross-track separation. Differential perturbations therefore motivate the need for formation-control strategies to preserve the desired formation geometry within mission constraints. Since stronger natural drift generally requires more frequent or larger corrective measures, differential perturbations also become direct drivers for formation maintenance fuel consumption, and thus, also mission lifetime [18, 20]. The following subsections will therefore summarize the most important differential perturbation effects for LEO orbiting formations.

### 4.5.1 Natural Relative Drift Under Ideal Two-Body Motion

For a leader-follower spacecraft pair, the relative orbit can be described by the difference between their respective classic orbital elements. If the leader has orbital elements $\boldsymbol{\alpha}_L$, and the follower has $\boldsymbol{\alpha}_F$, the differential orbital elements can be expressed as

$$\Delta\boldsymbol{\alpha} = \boldsymbol{\alpha}_F - \boldsymbol{\alpha}_L = \begin{bmatrix}\Delta a & \Delta e & \Delta i & \Delta\Omega & \Delta\omega & \Delta M_0\end{bmatrix}^T \tag{4.19}$$

where the components represent the differences in semimajor axis, eccentricity, inclination, RAAN, argument of perigee and mean anomaly at a chosen epoch, respectively. Under the ideal two-body assumption, where the Earth is modeled as a spherical body and all perturbing forces from Section 4.4 are neglected, the only form of secular formation drift is caused by a difference in the semimajor axis. This is stated by Alfriend et al. in [20], who explain that the drift only manifest in the along-track direction, resulting in continuous differential mean anomaly growth. This interpretation is also consistent with Hammerl & Schaub, who show that the relative radial offset must be zero for bounded relative motion in [21].

Other differential orbital elements still contribute to natural relative motion, but they do not cause secular drift in the ideal two-body problem. Instead, they induce bounded periodic motion. For example, a differential inclination produces periodic cross-track motion as the two orbital planes are tilted with respect to each other. Similarly, differential eccentricity contributes to in-plane radial and along-track oscillations. These periodic relative motions may be important for the formation geometry, but they do not themselves cause the formation to drift apart. However, additional secular drift mechanisms arise from differential perturbations, especially differential $J_2$ effects and atmospheric drag [22].

### 4.5.2 Differential $J_2$ Perturbations

As presented in Section 4.4.1, $J_2$ is the second degree, zeroth order zonal coefficient in the spherical harmonic representation of the Earth's gravity field. The coefficient represents the dominant term associated with the Earth's oblateness, and the zonal model of the same degree and order is therefore commonly known as the $J_2$ perturbation model. In a formation-flying context, the differential $J_2$ perturbation refers to the difference between the $J_2$-induced accelerations acting on the two spacecraft, and is one of the most important causes of secular formation drift in LEO.

The $J_2$ perturbing acceleration causes a slow precession of the orbit, seen primarily as precession of the orbital plane and a rotation in the argument of perigee. Alfriend et al. gives the corresponding per-satellite secular rates for RAAN and the argument of perigee in [20], which can be expanded to the following expression [23]

$$\dot{\Omega} = -\frac{3\bar{n}J_2R^2}{2a^2\left(1-e^2\right)^{1/8}}\cos i \tag{4.20}$$

$$\dot{\omega} = -\frac{3\bar{n}J_2R^2}{2a^2\left(1-e^2\right)^{1/8}}\left(\frac{5}{2}\sin^2 i - 2\right) \tag{4.21}$$

where $\bar{n}$ is the mean motion, $R$ is the Earth radius, and $a$, $e$ and $i$ are the satellite semimajor axis, eccentricity and inclination, respectively. Note that these per-satellite equations can be expressed as a function of each differential orbital elements by Taylor series expansion about the leader orbit, which has been derived for differential inclination in [20]. Examining Equations 4.20 and 4.21 shows that a difference in either semimajor axis, eccentricity or inclination between the satellites in a formation can cause $J_2$-induced secular rates, resulting in secular drift. This effect will manifest in the RTN frame as radial, cross- and along-track drift, depending on which secular rates differ. Because both secular rates are inversely proportional to $(1-e^2)$ a difference in eccentricity will cause much smaller secular rate than a corresponding difference in inclination for near-circular orbits where $e \approx 0$ [20].

### 4.5.3 Differential Atmospheric Drag

The atmospheric drag acceleration acting on a single spacecraft was introduced in Section 4.4.2. For formation-flying, however, the relative effect of drag is determined by the difference between the drag accelerations acting on each spacecraft. D'Amico & Montenbruck explain in [22], that atmospheric density variations can be neglected for close formations with sub-kilometer separation. Under this approximation, the main source of differential atmospheric drag is not a difference in local atmosphere, but a difference in the spacecraft ballistic coefficients. The ballistic coefficient is defined as

$$b = C_D\frac{A_D}{m_s} \tag{4.22}$$

where $C_D$, $A_D$ and $m_s$ are the drag coefficient, flow-normal cross-sectional area and spacecraft mass from equation 4.12. If the leader and follower spacecraft have ballistic coefficients $b_L$ and $b_F$, the relative ballistic coefficient difference can be written as $\varepsilon = (b_F - b_L)/b_L$. The resulting relative along-track drift over a time duration $\Delta t$ with a constant relative ballistic coefficient difference is given in [22]

$$\Delta r_{\hat{T}} = \frac{1}{2}\varepsilon||\mathbf{a}_D||_2 \cdot \Delta t^2 \tag{4.23}$$

where $||\mathbf{a}_D||_2$ denotes the drag acceleration magnitude of the leader spacecraft. This expression shows that small ballistic coefficient differences can accumulate into substantial along-track drift if the difference is persistent throughout significant time. Finally, it is important to note that the ballistic coefficient is attitude dependent though the flow-normal cross-sectional area, which makes relative spacecraft attitude an important driver for relative along-track drift and subsequent fuel consumption [20, 22].

## 4.6 GNSS-R Mission Concepts

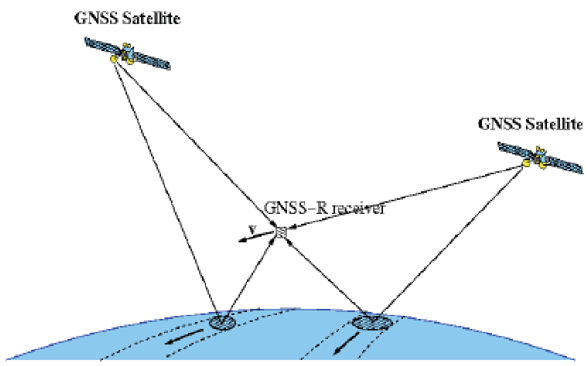


**Figure 4.6.1:** Illustration of the GNSS-R measurement principle. Courtesy of [24]

.

### 4.6.1 Remote Sensing Principles

GNSS-R is a passive remote sensing technique that uses reflected GNSS signals as signals of opportunity. As shown in Figure 4.6.1, GNSS-R forms a bistatic radar configuration where the GNSS satellite acts as a non-cooperative transmitter, while the observing spacecraft carries the receiver. Zavorotny et al. describes this concept in [25] as GNSS bistatic radar of opportunity, there the broadcasted GNSS signals illuminate the Earth's surface, and the receiver processes the scattered signal to infer surface properties such ass ocean wave height, wind speeds, soil moisture, etc. Hva skal jeg skrive he? Lykke til på gasskammeret Hva i alle dager skal jeg skrive? Jeg fortalte Vegard at jeg skulle LTFI, så nå får jeg faen meg gjøre det også

### 4.6.2 Cooperative GNSS-R Measurements

### 4.6.3 Operational Constraints

CHAPTER

# FIVE

# SHARED SIMULATOR CONFIGURATION

## 5.1 Shared Spacecraft Configuration

## 5.2 Shared Orbital Environment Models

## 5.3 Shared Numerical Integration Configuration

CHAPTER

# SIX

# COMPARATIVE ORBITAL PROPAGATOR

## 6.1 Simulation Architecture

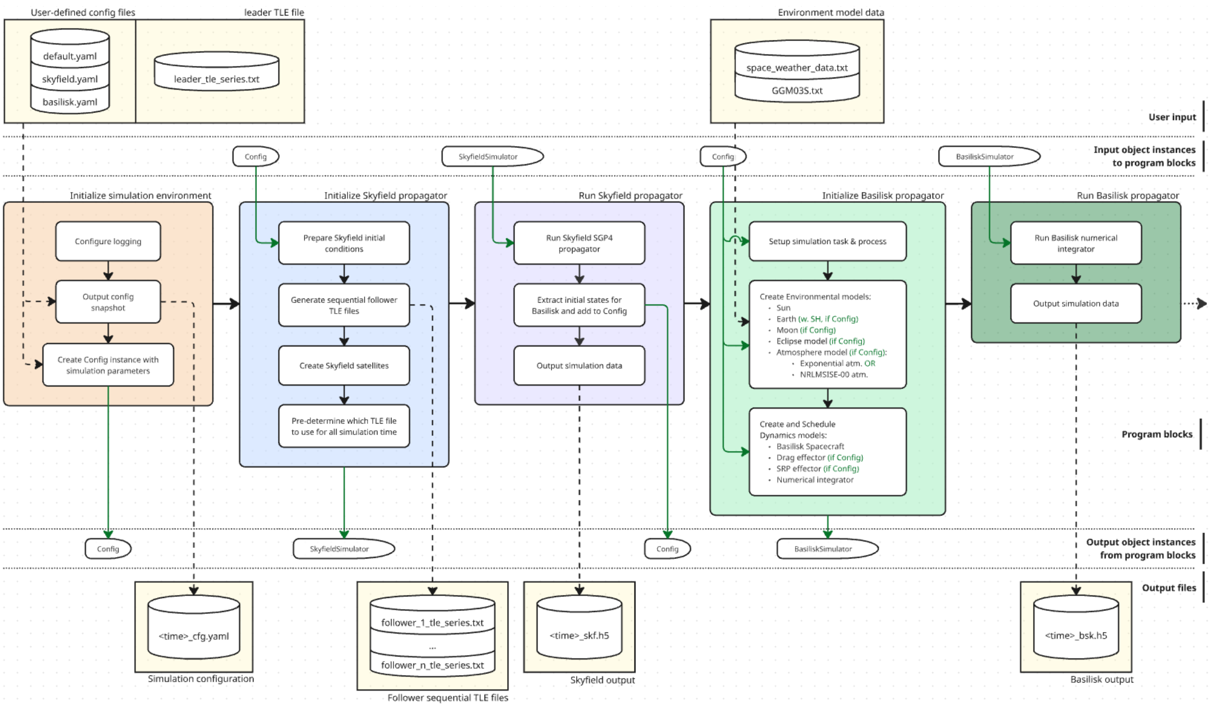


**Figure 6.1.1:** Adopted from [4]

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## 6.2 Skyfield SGP4 Reference Methodology

## 6.3 Propagation Initialization & Execution

CHAPTER

# SEVEN

# GNSS-R FORMATION FLIGHT SIMULATOR

## 7.1 Simulation Architecture

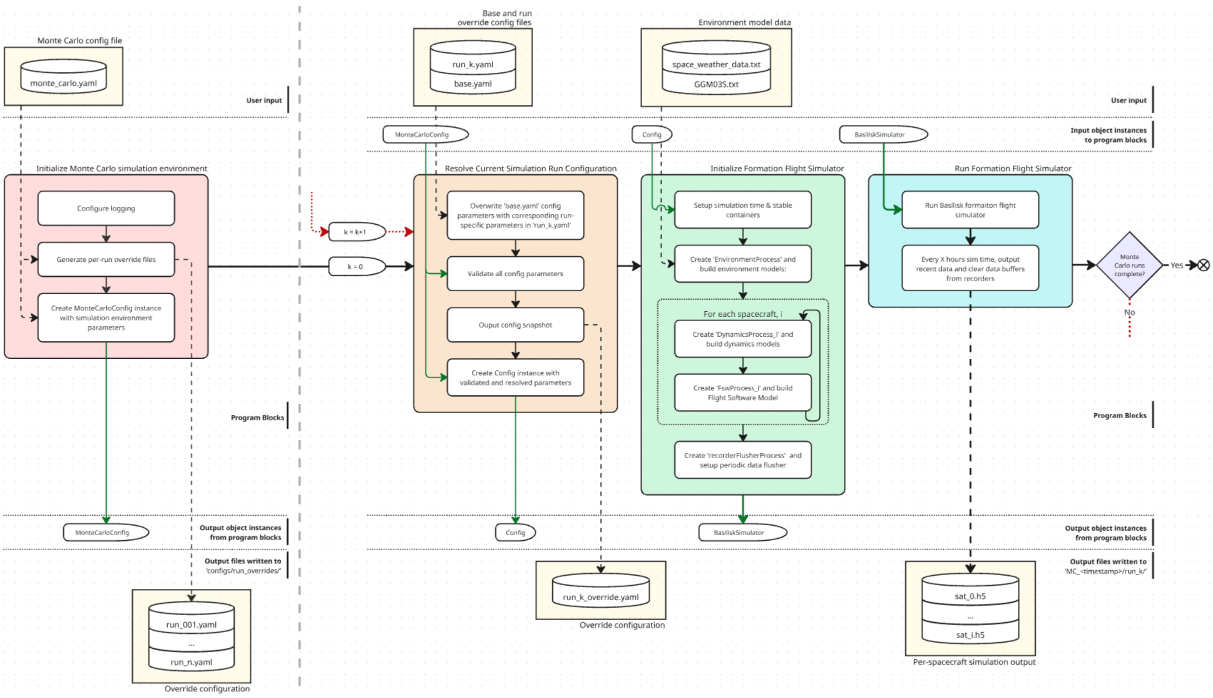


**Figure 7.1.1:** Formation flight simulator flowchart

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## 7.2 Electronic Power System

## 7.3 Flight Software

### 7.3.1 Software Architecture

### 7.3.2 Pointing Guidance Logic

### 7.3.3 MRP Feedback Pointing Controller

### 7.3.4 Formation Controller

## 7.4 Deployment Initialization

CHAPTER

# EIGHT

# VALIDATION & MISSION ANALYSIS METHODOLOGY

## 8.1 Propagation Validation Methodology

## 8.2 GNSS-R Mission Analysis Methodology

### 8.2.1 Mission Lifetime Definition

### 8.2.2 Evaluation Metrics

### 8.2.3 Monte Carlo & Scenarios

# CHAPTER NINE

# RESULTS

Plotting ideas for part 2:

- The big important plot: Fuel level over time
- It would be interesting to see natural formation drift, and calculate delta-V requirements to compensate for this drift to estimate lifetime. This estimate could be used to validate results.
- Show the relative differences in applied perturbation acceleration from differential J2 and drag to explain drift rate
- Each subsystem should be validated in addition to showing the full fuel consumption over simulation duration:
  - RW speed/torque step response
  - Thruster force output as a function of fuel level (/tank pressure)
  - MRP feedback controller tracking error and/or step response (would also be interesting to see a step response for different fuel levels)
  - Decomposed net power and battery level (should also show the illumination factor in the same/adjacent plot and the FSW state)

CHAPTER

# TEN

## DISCUSSION

*IMPORTANT*!

I have done some initial testing using the Basilisk Station Keeping example, and it seems like a strict CPO or Concentric formation is unachievable for the spacecraft reconfig module. If I have time, I should absolutely implement a HCW/ROE based controller for formation control. If I don't have time, I should just argue that this should be expanded in the future, and make a solid discussion about how my fuel consumption estimate is in comparison to the more realistic HCW case.

CHAPTER

# ELEVEN

# CONCLUSIONS

## 11.1 Conclusion

# REFERENCES


[1] Mariusz Eivind Grøtte et al. "Spacecraft Attitude and Angular Rate Tracking using Reaction Wheels and Magnetorquers". In: *IFAC-PapersOnLine* 53.2 (2020), pp. 14819–14826. DOI: 10.1016/j.ifacol.2020.12.1924.

[2] Gabriel Popescu. "Pixel Geolocation Algorithm for Satellite Scanner Data". In: (2014). PDF provided by user; publication year and journal not specified. URL: https://www.researchgate.net/publication/311807426_Pixel_geolocation_algorithm_for_satellite_scanner_data.

[3] Sascha Metz. "Implementation and Comparison of Data-Based Methods for Collision Avoidance in Satellite Operations". Master's Thesis. Darmstadt, Germany: Technische Universität Darmstadt, 2020. URL: https://www.researchgate.net/publication/344268107_Master_Thesis_Implementation_and_comparison_of_data-based_methods_for_collision_avoidance_in_satellite_operations.

[4] Christoffer D. Hystad. *Comparison of Skyfield SGP4 and Basilisk Numerical Orbit Propagation for GNSS-R Formation-Flying Applications*. Specialization Project Report. Norwegian University of Science and Technology (NTNU), Department of Engineering Cybernetics, Dec. 2025.

[5] Olav Egeland and Jan Tommy Gravdahl. *Modeling and Simulation for Automatic Control*. Trondheim, Norway: Marine Cybernetics, 2002. ISBN: 82-92356-01-0.

[6] NASA Goddard Space Flight Center. *Appendix C: Description of Selected Coordinate Systems Used in SSC Programs*. https://sscweb.gsfc.nasa.gov/users_guide/Appendix_C.html. Satellite Situation Center (SSCWeb) User Guide, accessed 2026-05-10. 2025.

[7] John Stryjewski. *Coordinate Transforms*. Aug. 2020. URL: https://x-lumin.com/wp-content/uploads/2020/09/Coordinate_Transforms.pdf.

[8] David A. Vallado. *Fundamentals of Astrodynamics and Applications*. 2nd ed. Vol. 12. Space Technology Library. El Segundo, CA: Microcosm Press and Kluwer Academic Publishers, 2001. ISBN: 9781881883128.

[9] Sean R. Stratton. "Benefits of RTN Reference Frame as Standard for Optical Observation Data". In: *Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS)*. Approved for Public Release, Distribution Unlimited. SciTec, Inc. 2025. URL: https://amostech.com/TechnicalPapers/2025/Astrodynamics/Stratton.pdf.

[10] O. Montenbruck et al. "GNSS satellite geometry and attitude models". In: *Advances in Space Research* 56.6 (2015), pp. 1015–1029. ISSN: 0273-1177. DOI: 10.1016/j.asr.2015.06.019.

[11] Thor I. Fossen. *Handbook of Marine Craft Hydrodynamics and Motion Control*. 2nd ed. Chichester, UK: John Wiley & Sons, 2021. ISBN: 978-1-119-57505-4. DOI: 10.1002/9781119994138.

[12] Hanspeter Schaub and John L. Junkins. *Analytical Mechanics of Space Systems*. 4th ed. AIAA Education Series. Reston, VA: American Institute of Aeronautics and Astronautics, 2018. ISBN: 978-1-62410-521-0. DOI: 10.2514/4.105210.

[13] Malcolm D. Shuster. "A Survey of Attitude Representations". In: *The Journal of the Astronautical Sciences* 41.4 (Oct. 1993), pp. 439–517.

[14] Stephen A. O'Keefe and Hanspeter Schaub. "Shadow Set Considerations for Modified Rodrigues Parameter Attitude Filtering". In: *Journal of Guidance, Control, and Dynamics* (2014). DOI: 10.2514/1.G000405.

[15] S. R. Marandi and V. J. Modi. "A Preferred Coordinate System and the Associated Orientation Representation in Attitude Dynamics". In: *Acta Astronautica* 15.11 (1987), pp. 833–843. ISSN: 0094-5767. DOI: 10.1016/0094-5765(87)90042-5.

[16] Howard D. Curtis. *Orbital Mechanics for Engineering Students*. Elsevier Aerospace Engineering Series. Oxford, UK: Elsevier Butterworth-Heinemann, 2005. ISBN: 0-7506-6169-0. URL: https://www.hlevkin.com/hlevkin/90MathPhysBioBooks/Mechanics/Curtis_OrbitamMechForEngineeringStudents.pdf.

[17] Astronomical Returns. *Section 4.3 – The Six Orbital Elements*. Accessed online. n.d. URL: https://www.astronomicalreturns.com/p/section-43-six-orbital-elements.html (visited on 05/12/2026).

[18] Oliver Montenbruck and Eberhard Gill. *Satellite Orbits: Models, Methods, and Applications*. Berlin, Heidelberg: Springer, 2000. ISBN: 978-3-540-67280-7. DOI: 10.1007/978-3-642-58351-3.

[19] J. M. Picone et al. "NRLMSISE-00 empirical model of the atmosphere: Statistical comparisons and scientific issues". In: *Journal of Geophysical Research: Space Physics* 107.A12 (2002), p. 1468. DOI: 10.1029/2002JA009430.

[20] Kyle T. Alfriend et al. *Spacecraft Formation Flying: Dynamics, Control and Navigation*. Elsevier Astrodynamics Series. Oxford, United Kingdom: Butterworth-Heinemann, 2010. ISBN: 978-0-75-068533-7.

[21] Julian Hammerl and Hanspeter Schaub. "Orbiting Spacecraft Relative Motion in the Inertial Frame". In: *The Journal of the Astronautical Sciences* 73.21 (2026). DOI: 10.1007/s40295-025-00556-w.

[22] Simone D'Amico and Oliver Montenbruck. "Proximity Operations of Formation-Flying Spacecraft Using an Eccentricity/Inclination Vector Separation". In: *Journal of Guidance, Control, and Dynamics* 29.3 (May 2006), pp. 554–563. DOI: 10.2514/1.15114.

[23] Blake B. Hajovsky. "Satellite Formation Control Using Atmospheric Drag". Master's thesis. Wright-Patterson Air Force Base, Ohio: Air Force Institute of Technology, Mar. 2007. URL: https://scholar.afit.edu/etd/2989/.

[24] European Space Agency (ESA). *Earth Sciences: Remote Sensing Using GNSS-R*. 2013. URL: https://gssc.esa.int/navipedia/index.php/Earth_Sciences.

[25] Valery U. Zavorotny et al. "Tutorial on Remote Sensing Using GNSS Bistatic Radar of Opportunity". In: *IEEE Geoscience and Remote Sensing Magazine* (Jan. 2015), pp. 8–45. DOI: 10.1109/MGRS.2014.2374220.

# APPENDICES

# A - GITHUB REPOSITORY

All simulator code and LaTeX-files used in this document are included in the GitHub repositories linked below.

## Specialization project report LaTeX repository link

- https://github.com/chystad/Specialization-Project-Report-LaTeX-

## Orbit Simulator repository link

- https://github.com/chystad/Orbit_Simulator

# B - NUMERICAL VALUES

## Numerical Values for Physical Constants

| $Constant$ | $Value$ | $Unit$ |
|---|---|---|
| $J_0$ | $-1$ | - |
| $J_1$ | $0$ | - |
| $J_2$ | $1.08263 \times 10^{-3}$ | - |
| $J_3$ | $-2.53266 \times 10^{-6}$ | - |
| $J_4$ | $-1.61962 \times 10^{-6}$ | - |
| $P_{Sun}$ | $4.578299965 \times 10^{-6}$ | $Ws/m^3$ |
| $1AU$ | $149597871$ | $km$ |
| $R$ | $6378136.6$ | $m$ |
| $H_0$ | $7200$ | $m$ |
| $\rho_0$ | $1.225$ | $kg/m^3$ |
| $-$ | $-$ | $-$ |
| $-$ | $-$ | $-$ |
| $-$ | $-$ | $-$ |

**Table .0.1:** Physical constants and associated numerical value. $J$-terms courtesy of